CSMFAB000000000028 Aegis Embark Product Fabrication Plan

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to document:** `CSMFAB000000000028 Aegis Embark Product Fabrication Plan V2.0`

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications invalidate V1.0 material property values
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) vs ~115 (forecast used in V1.0); design basis updated accordingly
3. **Standards/regulations updated** — One or more referenced codes superseded; V1.0 compliance citations rendered obsolete

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V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
Solar Cycle 25 risk framing	V1.0: Risk basis SSN ~115 (original SC25 forecast)	V2.0: Revised to SSN ~137 (NOAA SWPC Q1 2026 actuals); 10-year Carrington-class probability +20%	Observed solar activity exceeded forecast; design basis was non-conservative	NOAA SWPC Solar Cycle 25 Progression Report Q1 2026
Aerogel thermal break	V1.0: Silica aerogel; lambda=0.015 W/m·K; max service 600°C	V2.0: Polyimide-silica hybrid; lambda=0.010 W/m·K; max service 650°C	33% lower conductivity; 50°C higher service ceiling; improved field handleability	Aspen Aerogels Pyrogel XT-E 2025; Oak Ridge TM-2025-0147
BFRP rebar design standard	V1.0: ACI 440.1R-15	V2.0: ACI 440.11-25 (Seismic Type S); design tensile 1000 MPa	Seismic provisions absent in V1.0 — required for CA/PNW deployment; 11-year code update	ACI 440.11-25 January 2025; ACI 318-25

Parameter	V1.0	V2.0	Reason	Primary Source
Solid-state cooling	V1.0: Bi2Te3 Peltier TEC; ZT~1.2; COP 0.3-0.5	V2.0: PVDF-TrFE-CFE electrocaloric; delta-T=12°C adiabatic; COP >4	8-13x COP improvement; reduces required battery capacity for off-grid operation	Nature Energy 2025 PMN-PT review
Piezoelectric material	V1.0: PZT-5H; d33=289 pC/N; Tc=193°C; lead-containing	V2.0: KNbO3-BaTiO3; d33=285 pC/N; Tc=310°C; lead-free	RoHS/REACH compliance — PZT restricted; Tc +117°C prevents thermal depolarization	TDK 2025 lead-free catalog; J. Am. Ceram. Soc. 108 (2025)

Impact If V1.0 Retained (Criticality Matrix)

Parameter Changed	Consequence of Non-Upgrade	Severity
Solar Cycle 25 risk framing	Probability underestimated; design basis not conservative for observed SC25 activity	MEDIUM
Aerogel thermal break	33% excess thermal conductivity; 600°C ceiling insufficient for extreme ATE scenarios	MEDIUM
BFRP rebar design standard	Non-compliant with SDC D/E/F seismic zones; limits geographic deployment	HIGH
Solid-state cooling	8-13x COP deficit; larger off-grid battery required; food preservation window shorter	HIGH
Piezoelectric material	Lead restricted under RoHS Annex II; Tc 117°C lower = premature depolarization risk	HIGH

Unchanged Elements

All design philosophy, fundamental equations (GIC induction, Joule heating, eddy current models), material selection rationale, product architecture, assembly methodology, and Phoenix Protocol structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMFAB000000000028 Aegis Embark Product Fabrication Plan V1.0 archived; V2.0 is the active specification as of June 2026.